HAPLight v3

Research & Development Plan

Status audit, literature positioning, new contributions, experimental protocol, compute plan and repository blueprint

Project	HAPLight: heterogeneity-aware occupancy-pressure signal control with certified reinforcement learning for Indian mixed traffic
Author	Aditya Singh (RGIPT)
Version / date	v1.0, 4 November 2026
Supersedes	`PLAN.md` (Aug 2026) as the forward plan. `PAPER_CHANGELOG.md` remains the historical evidence log.
Inputs audited	Full repository (src, scripts, tests, configs, nets, results*), `PAPER_CHANGELOG.md` (4,260 lines, entries C-1 . . . P5-7), `paper/HAPLight_journal_v2.tex`, the machine, and the 2023–2026 literature.
Status	Approved 4 Nov 2026. The decisions taken are recorded in Sec. 15.

One-paragraph verdict

The previous development effort was rigorous: it found and fixed several serious defects, most importantly the *8× vehicle double-counting bug* (P5-2), which invalidated every model and absolute queue metric produced before 9 September 2026. On corrected measurements, the learned controller (HAPLight-RL) beats greedy HAP by 1.7–4.5% median travel time in all four (scenario, noise) cells, with a significant BCa CI on ATT, AWT and PWD in every cell. The rest of the paper's claims are thin: greedy HAP is indistinguishable from count-based max-pressure on travel time; the learned-baseline set collapsed (two diverged, one crashed mid-retrain, two never trained); the Bengaluru scenario was cut; there is one training seed; and the stability "certificate" assumes infinite link storage, which the project itself showed to be false in practice. In October 2026, a close competitor appeared (Com-MP, arXiv:2610.31037). It proves stability for *any* RL output under *finite* storage. The paper therefore needs a new centre of gravity. This plan proposes four. (i) A **Certified Action Set** (CAS): RL may only choose phases within a tunable slack of the max-pressure optimum, which gives a provable guarantee with no runtime override. (ii) **Storage-aware** HAP with a finite-storage analysis and a statistically powered deadlock study. (iii) **Real-data grounding** of the sensing model on Bengaluru CCTV images (UVH-26), the first use of the GPU. (iv) A **reviewer-proof evaluation**: fidelity-gated baselines including the 2026 competitor, ≥3 training seeds, a real-geometry Bengaluru corridor, and a clean reproducible repository. Estimated time to submission: about 12 weeks.

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1 Executive summary

Where we are. Phases 0–4c and most of 5 of the August plan are done. Phase 6 (Bengaluru) was cut. Phases 7–9 (full sweep, reporting, paper rewrite) have not started. The last activity (13 Sep 2026) was a corrected-measurement CoLight retrain on S1. It crashed at iteration 24/40 with a TraCI socket error, and the follow-up chain (`scripts/_phase5_retrain4.sh`) was written but never launched. The paper source has not been edited, by design. The repository is **not under version control**.

What is solid. The simulation stack (SUMO 1.27.1, SL2015 sublane with validated two-wheeler filtering), the corrected measurement pipeline with a ground-truth gate, the episode-isolation fixes, the reward validation protocol, and the 84-test suite all hold up; the suite passed during this audit in 16 min. Corrected HAPLight-RL checkpoints exist for S1 and S2 (40 PPO iterations, seed 0).

What must change.

1. **Contribution.** Replace "RL + queue-threshold safety filter" (Corollary 1, infinite buffers) with a *Certified Action Set*. This gives a structural guarantee for any policy, composes with sensing error, has no override discontinuity, and leaves no masked-gradient steps.
2. **Finite storage.** Promote the project's own deadlock discovery to a contribution: storage-aware HAP pressure, a finite-storage proposition or theorem, and a *powered* deadlock-rate experiment ($n \approx 200$ paired episodes per cell).
3. **Real data.** Use UVH-26 (26,646 Bengaluru CCTV frames, 14 Indian classes, CC-BY-4.0) to (a) cite the fleet mix, (b) calibrate the detector-noise model from real confusion matrices, and (c) test Lemma 1 on real frames.
4. **Evaluation.** Use fidelity-gated learned baselines (including a Com-MP adaptation and finite-storage max-pressure baselines), $\geq$3 training seeds, 20 evaluation seeds per cell, hierarchical bootstrap statistics, and a rebuilt real-geometry Bengaluru corridor.
5. **Engineering.** Create a new git repository outside OneDrive with bulk data on `D:`, use the libsumo backend, a single locked environment, CI, and a session status file so that a lost session never again loses context.

Timeline. About 12 weeks: infrastructure and re-validation (weeks 1–2), method, theory and baselines (weeks 2–5), compute sweep (weeks 6–9), statistics and paper (weeks 10–12). Target: arXiv plus submission to IEEE T-ITS or Transportation Research Part C in early February 2027.

2 Status audit: what has been done and what it achieved

2.1 Phase-by-phase status

#	Phase	Status	What was achieved / caveats (changelog entry)
0	Setup, SUMO, tests	**DONE**	SUMO 1.27.1 binary. **Caveat:** training ran on *system* Python 3.10 with `traci 1.25` over TCP. The `.venv` has `libsumo 1.27.1` and was never used, so the client/server version mismatch and the slow socket backend are both live (Sec. 3).
1	Env, classical controllers, metrics, S1/S2 nets	**DONE**	FT (Webster on true demand), SOTL, MP, Efficient-MP, Advanced-MP (corrected to the Zhang et al. stay/switch rule, P4c-16), GreedyHAP. Gate "MP beats FT" passes only after the 8× fix (P5-2).
1b	Demand calibration	**DONE**	Capacity S1 4,482 veh/h, S2 6,543 veh/h; nominal v/c≈0.85, surge ≈1.05 (P1c-5). Measured *before* the 8× fix; throughput-based, so probably unaffected, but must be re-verified.

#	Phase	Status	What was achieved / caveats (changelog entry)
1c	Per-movement $\bar{a}_m$, fleet mix, H1 evidence	**PARTIAL**	Per-movement commissioning calibration closes the estimator gap to 0.16–0.57% (P1c-1). The fleet mix (2W 62 / car 22 / auto 10 / bus 2 / truck 4) is **uncited**. H1 rests on 3 seeds and is pre-fix.
2	C1/C2 calibration (Lemma 1, A5)	**PARTIAL**	HCM vs area: total error 0.105 vs 0.113. The margin is only 7%, from a **single unseeded, pre-fix** run (P1c-10). Fitted 2W discharge equivalency 0.491, close to Indo-HCM's 0.5.
3	Two-wheeler sublane	**DONE**	2W finish 7.6–12 ranks ahead of cars under SL2015 vs 0.35–3.9 under LC2013 (P3-1/P3-3). The sublane cost on a full episode is only 1.19×.
4	Policy, PPO, filter	**DONE**	Paper Eq. 12 reward was **wrong-signed** (random policy scored highest). Replaced by `total_queue`, which passes the validity gate (P4-4). 23,650-parameter policy verified.
4c	Storage-aware HAP, deadlock	**DONE**	Finite-storage deadlock discovered (worst movement 377–715% of storage). Edge-level barrier failed; movement-resolution barrier plus storage-triggered filter plus hysteresis works. Threshold re-swept after the fix to 0.15/0.10 (P5-2).
5	RL baselines	**PARTIAL**	Code exists for MPLight, Advanced-MPLight, PressLight, IPPO and CoLight-variant (single-hop). After the fix, `pressure_raw` **fails** the reward gate on S1, and MPLight and Advanced-MPLight diverge, so they were dropped from the laptop table (author decision P5-7). The CoLight S1 retrain crashed at iteration 24/40. PressLight and IPPO were never trained.
6	Bengaluru S3	**CUT**	5 valid signals extracted. 62% of arterial edges are <40 m single-lane, so the corridor saturates at the lowest demand. Widening join-dist made it worse. Cut permanently (Phase-6 entries).
7	Full sweep	**NOT STARTED**	No robustness grid, ablations, sensitivity, fairness, surge or weight-vector (H5) runs.
8	`report.py`, `stats.py`	**NOT STARTED**	Statistics live inside `scripts/evaluate_policy.py` (BCa, rank-biserial, Wilson). There is no LaTeX table generator.
9	Paper rewrite	**NOT STARTED**	`.tex` untouched; all placeholders open.

2.2 The latest valid evidence (corrected measurements)

Only results produced after the P5-2 fix are valid. Checkpoints are `results/s1_seed0_1788963312` and `results/s2_seed0_1788987398` (HAPLight-RL, 40 iterations, seed 0, production filter 0.15/0.10). The table below is the P5-5 preliminary evaluation: medians over $n = 10$ paired demand seeds.

Median ATT (s) / AQL / gridlock	S1, $\eta = 0$	S1, $\eta = 0.6$	S2, $\eta = 0$	S2, $\eta = 0.6$
FixedTime	237.8 / 134.6 / 0	237.8 / 134.6 / 0	233.5 / 214.5 / 1	233.5 / 214.5 / 1
MaxPressure	243.6 / 138.4 / 0	240.4 / 140.9 / 2	215.0 / 155.9 / 0	218.2 / 184.6 / 1
GreedyHAP	239.2 / 129.0 / 0	238.9 / 127.3 / 0	215.8 / 155.6 / 1	216.1 / 156.8 / 0
HAPLight-RL	**230.9 / 113.6** / 0	234.8 / 119.2 / 0	206.1 / 126.5 / 0	207.3 / 132.7 / 0
HAPLight-RL (no filter)	234.5 / 118.1 / 0	233.0 / 118.3 / 1	204.9 / 126.5 / 1	206.2 / 130.5 / 1

Reading.

- **HAPLight-RL vs greedy HAP.** ATT is 3.5%, 1.7%, 4.5% and 4.1% lower. The ATT, AWT and PWD BCa CIs exclude zero in all four cells; AQL in three of four.
- **Greedy HAP vs MP.** ATT differs by −0.4% to +1.8%, i.e. *no practical difference*. H1, the paper's first claim, is not supported on travel time.
- **Deadlock.** Filtered 0/40 vs unfiltered 3/40 (McNemar $p = 0.25$). The direction is right but not significant, and two of the three events share one demand seed.

- **Caveat on AQL.** AQL is measured in HAP's own weight units (Indo-HCM PCU), which favours HAP. The vehicle-count AQL must also be reported.

2.3 Evidence ledger: valid, retired, excluded

- **Valid:** everything above; P3 sublane results; P4-4 reward-gate logic and the P5-4 re-run; P1c-1 per-movement calibration mechanism; all code fixes.
- **Retired (marked `INVALID_8X.txt`):** every checkpoint and absolute AQL/occupancy number before 9 Sep 2026, including the Phase 4 100-iteration models, the Phase 5 40-iteration table and the "0% vs 5–10% gridlock" headline, the P4c-9 to P4c-14 threshold sweeps (0.80), and the P4-6 $\eta = 0.6$ table.
- **Excluded (marked `EXCLUDED_PRESSURE_RAW.txt`):** MPLight and Advanced-MPLight trained on `pressure_raw`.
- **Needs re-verification:** C1/C2, the capacity scales, `a_bar_per_movement.json`, and H1 spill-back evidence. All were measured before the fix. Their dependence on the count channel is mostly ratio-based, but it was never checked.

3 System and environment audit

Item	Found	Implication / recommendation
CPU	Intel i7-11800H, 8 cores / 16 threads	Fine for SUMO. SUMO is single-threaded per instance, so parallelism comes from processes.
RAM	**7.7 GB**, one 8 GB DIMM (ChannelA-DIMM0); 0.74 GB free at audit time	**The binding constraint.** 15 parallel SUMO instances caused a MemoryError (P4c-15). The laptop very likely has a second slot (verify the model). Upgrading to 16–32 GB is the cheapest large speed-up, enabling 12–14 parallel environments instead of 6–8.
GPU	GTX 1650 Max-Q, 4 GB, driver 610.62 (CUDA UMD 13.3); **never used**	Both torch builds are CPU-only. The GPU is not useful for SUMO-bound RL training, since the policy is tiny. It *is* the right tool for the new vision track (UVH-26 detector inference) in a separate CUDA environment. Check that the chosen wheel supports sm_75 (Turing).
Disk	C: NVMe, 25 GB free of 477; D: Seagate USB HDD, 742 GB free	Code and the virtual environment on C: (outside OneDrive). Runs, checkpoints, logs and datasets on `D:\haplight-data`.
OneDrive	Project lives in the synced folder	Measured ≈2.2 cores of live contention during training (P5-6). Move the working repository out of OneDrive.
Python	System 3.10.0 (torch 2.9.1+cpu, **traci/sumolib 1.25**, no libsumo); `.venv` 3.10.0 (eclipse-sumo/libsumo/traci 1.27.1, torch 2.13+cpu)	**Two divergent environments**, and the training scripts call the wrong one. Python 3.10 reached end-of-life in Oct 2026. Use one `uv`-locked Python 3.12 environment with `eclipse-sumo==libsumo==traci==1.27.1`.
SUMO backend	TraCI over TCP in all runs	libsumo is typically 2–3× faster and has no sockets, which also removes the crash class seen on 13 Sep. Requires a backend-parity test (Sec. 12).
Tooling	uv 0.13.1, git 2.49, MiKTeX (pdflatex, latexmk), WSL2 Ubuntu-24.04 (stopped); no `gh`, no Docker	Install `gh` for the GitHub remote. WSL2 is an optional fallback if Windows libsumo misbehaves.
Version control	**None** (not a git repository)	Initialise immediately: snapshot the current state as the v2 archive, then build v3 in a new repository.
Tests	84 passed in 16 min (system Python)	Green baseline to port.

4 Critical assessment of the v2 paper

We do not need to follow v2. The table judges each claim by evidence and by how a T-ITS or TR-C reviewer would read it.

Claim in v2	Status after audit	Action for v3
Lemma 1 (tight area-estimator bound, $\varepsilon^\star = 0.450$ HCM / 0 area)	Correct but elementary: it is a weighted-mean bound. Not validated on real images.	Keep as a design rule. Add a **real-image validation on UVH-26** and per-movement commissioning (P1c-1).
Theorem 1 (throughput optimality of weighted MP)	Varaiya's argument with positive weights. Weighted or priority MP retaining stability is already known (e.g. OCC-MP 2024). The slotted model ignores clearance loss, yet $\Delta t = t_y = 5$ s here.	Restate honestly as an extension. Add a frame-based treatment of switching loss (cf. VT-MP 2026). Novelty comes from elsewhere.

Claim in v2	Status after audit	Action for v3
A5 (composition-invariant saturation flow) tested in C2	Interesting: HCM satisfies A5 better than area (CV 1.9% vs 7.5%), while area is exact for sensing. A genuine trade-off, but from a single unseeded pre-fix run.	Re-run with ≥5 seeds and CIs. Present H5 as a trade-off result.
Prop. 1 (dispersion, not magnitude, of sensing error costs capacity)	Consistent with approximate max-weight theory. Charalambous et al. (IFAC 2023) show max-weight stays throughput-optimal under noisy queues.	Cite and narrow the claim to the multiplicative per-movement characterisation. Combine with A5 violation into one additive error budget (changelog C-3).
Reward = "exact Lyapunov drift functional" (Eq. 12)	**False**: the implemented reward is anti-aligned (P4-4).	Reward is `total_queue`. The theory link becomes "reward = the quantity the stability bound controls".
Corollary 1 (filter gives a certified floor)	Holds only for infinite buffers. Deadlock exists in practice. The benefit is directional only ($p = 0.25$). The discontinuous override caused chattering and masks policy-gradient steps. Com-MP (2026) offers a structural guarantee under finite storage.	**Replace with the Certified Action Set (Sec. 6).** Keep the filter as an ablation.
"Twelve baselines"	Delivered: 3 classical + 1 partially trained learned baseline.	Fidelity-gated baseline suite (Sec. 9).
5 seeds × 30 episodes	1 training seed; 10 evaluation seeds.	≥3 (target 5) training seeds × 20 evaluation seeds per cell, with hierarchical bootstrap.
OSM Bengaluru corridor	Cut.	Rebuild as an OSM-informed, real-geometry corridor (Sec. 8).
"First systematic sensing-robustness study"	Not run.	η grid including out-of-distribution levels, plus a **real-calibrated** noise condition.
Representative Indian fleet mix	Uncited.	Derive from UVH-26 class counts and cite.

5 Literature review and positioning (2019–Nov 2026)

5.1 What exists

Max-pressure foundations. Tassiulas & Ephremides (1992), Varaiya (2013), Le et al. (2015). Practical variants include cyclic MP (Levin 2020), switching-loss-aware learned MP (Wang et al., TR-C 2022) and delay-based MP (Liu & Gayah 2022, 2023).

Weighted / priority MP. OCC-MP for transit and high-occupancy vehicles (Ahmed, Liu & Gayah, TR-C 2024), pedestrian MP (Liu, Gayah & Levin, TR-C 2024), Transit-MP (Tan et al., 2025), Private-MP (Tan et al., TR-C 2026), and PCU-weighted queue MP for heavy-vehicle port networks (*Systems* 2026). These preserve stability under person or class *priorities*. None treats road-space heterogeneity, the validity of A5, or a classification-free camera estimator.

Finite storage / spillback. Capacity-aware back-pressure (Gregoire et al., IEEE TCNS 2015), the pressure-releasing policy with a stability proof on a reduced region (Xiao et al., CDC 2014), position-weighted back-pressure (Li & Jabari, TR-B 2019) and Hao et al. (2020). **PASCAL** (TR-C 2026) states that a formal stability proof under realistic finite-capacity spillback "remains an important avenue for future research". **VT-MP** (Tan, arXiv:2610.32280, Oct 2026) keeps infinite-storage theory and adds a heuristic receiving-supply gate.

Imperfect information. Noisy-queue max-weight (Charalambous et al., IFAC 2023; Liaquat, IET ITS 2024) and CV-MP with partial connected-vehicle observation (Tan et al., TR-B 2026).

Stability-preserving RL. Com-MP (Zhu & Tan, arXiv:2610.31037, 22 Oct 2026) is the closest

competitor. A shared MAPPO actor outputs a mixing weight $\alpha \in \{0, .25, .5, .75, 1\}$ between two normalised MP pressures (queue-MP and CV-MP); the phase is always the argmax of the mix. Strong stability is proved for *any* α sequence under *finite storage* (following Li & Jabari). It reports a 15.6% delay reduction vs Q-MP on a 45-signal Qinzhou network. *Not covered:* vehicle classes or PCU, sensing error in the theory, RL or fixed-time baselines, significance tests. Related: Reference RL (Lu et al., Inf. Sci. 2025), safe RL for transit priority (Zhou et al., ESWA 2025), SafeLight (2022, collision safety).

Learned TSC state of the art. PressLight, CoLight (2019), MPLight (2020), Advanced-XLight (ICML 2022), Unicorn (generalisable MARL, arXiv 2025, code public), and LLM controllers LLMLight (2024/25), Traffic-R1 and ProcessLight (2026). Benchmarks: RESCO (NeurIPS 2021), LibSignal (MLJ 2024), PyTSC (Sensors 2025), T-REX (2025) and Sim2Signal (2026). RESCO warns that reported gains often vanish under rigorous protocols.

Indian heterogeneous traffic. The hybrid MP-plus-RL controller of Agarwal et al. (ESWA 2024) has no stability analysis and no sensing model. Area-occupancy PCU work comes from Mallikarjuna & Rao and Chandra et al. (J. Transp. Eng. 2018). Several YOLO-plus-RL application papers exist (2025) without guarantees.

Data. UVH-26 (IISc AIM with Bengaluru Traffic Police, arXiv:2511.02563, Nov 2025): 26,646 1080p CCTV frames from ≈2,800 Bengaluru cameras, ≈1.8 M boxes, 14 classes (two-wheeler, three-wheeler, hatchback, sedan, SUV, MUV, bus, truck, LCV, mini-bus, tempo-traveller, bicycle, van, other), COCO format, CC-BY-4.0. Detectors are released under Apache-2.0. There are no camera IDs in the annotation files.

5.2 Gap analysis and novelty claim

Work	Veh. het.	Stability	Fin. stor.	Sens. theory	RL	Real data	India
Varaiya 2013 / weighted MP (OCC-MP 2024)	– / priority	✓	–	–	–	–	–
CABP 2015, PWBP 2019, PASCAL 2026	–	✓/ partial	✓	–	–	–	–
CV-MP 2026, noisy MW 2023	–	✓	–	partial	–	–	–
Com-MP 2026	–	✓	✓	–	✓	network	–
Advanced-XLight 2022, Unicorn 2025	–	–	–	–	✓	flows	–
Agarwal et al. 2024	✓	–	–	–	✓	–	✓
HAPLight v3 (proposed)	✓	✓ (CAS)	✓	✓	✓	UVH-26	✓

Defensible novelty statement for v3

HAPLight v3 is, to our knowledge, the first signal-control framework that combines: (i) road-space heterogeneity-weighted pressure with an *empirically tested* saturation-flow assumption; (ii) a classification-free camera estimator whose error bound is validated on *real Indian CCTV data*; (iii) a *Certified Action Set* under which *any* learned policy inherits throughput optimality, with explicit composition of sensing and saturation-flow errors; (iv) storage-aware pressure with a finite-storage analysis and a statistically powered deadlock study; and (v) an Indian-calibrated, fidelity-gated, multi-seed evaluation. Com-MP must be cited, discussed and used as a baseline. Our differentiators are (i), (ii) and the error-composition part of (iii), none of which Com-MP addresses.

6 Proposed direction: HAPLight v3 contributions

Every contribution must end as either a proved statement (with a numerical check) or a pre-registered experiment. Speculation does not enter the paper.

C1. HAP measure and weighted max-pressure (retained, re-attributed)

Keep Eqs. (1)–(9) of v2. Restate Theorem 1 as a *weighted extension* of Varaiya, proved for any $\rho > 0$ under A5. Add switching and clearance loss through a frame argument: the guaranteed region shrinks to $(1 - I/T)\Lambda$ for frame length T, in the style of VT-MP Prop. 1 and Le et al. (2015). **Proof**

obligation P1: frame-based drift with $\Delta t = t_y$.

C2. Classification-free estimator, grounded in real data

Keep Lemma 1. Add (a) per-movement commissioning calibration (P1c-1 finding) and (b) validation on UVH-26 (Sec. 7). The weight-vector choice becomes an honest two-criterion trade-off: estimator exactness (area wins) vs saturation-flow invariance A5 (HCM wins). Report $\varepsilon + \nu$ for both, with CIs.

C3. NEW — Certified Action Set (CAS) for learned control

Definition 1 (Certified action set)**.** *For slack parameters $\gamma \in [0, 1)$ and $C \geq 0$, let $\hat{P}_i^\star(t) = \max_\phi \hat{P}_i(\phi, t)$ be the maximum (storage-aware) HAP pressure computed from* sensed *queues. The certified action set at intersection i is*

$$\mathcal{A}_i^{\gamma,C}(t) = \Big\{\phi \in \Phi_i : \ \hat{P}_i(\phi, t) \ \geq \ \hat{P}_i^\star(t) - \gamma\,[\hat{P}_i^\star(t)]^+ - C\Big\}.$$

The argmax always belongs to it. The RL policy's logits are masked to $\mathcal{A}_i^{\gamma,C}(t) \cap \mathcal{M}_i(t)$, where $\mathcal{M}_i$ is the $g_{\min}/g_{\max}$ mask.

Theorem 2 (Target statement — stability of any CAS-constrained policy)**.** *Under A1–A5 and exact sensing, every policy (stochastic, history-dependent, e.g. a neural network) that selects $\phi_i(t) \in \mathcal{A}_i^{\gamma,C}(t)$ for all i, t makes the weighted queue process strongly stable for every λ with $\lambda/(1-\gamma) \in \operatorname{int}\Lambda$. The time-average queue bound grows linearly in $NC\Delta t$. Under per-movement multiplicative sensing error ε (Prop. 1) and A5 violation ν (changelog C-3), the same holds when the interiority margin exceeds $c_1\Delta t(\varepsilon + \nu)$.*

Proof route. This is a $(1-\gamma)$-multiplicative / C-additive approximate max-weight argument (Neely 2010; Tassiulas–Ephremides) applied to the separable functional of Appendix A. With $\gamma = C = 0$ it reduces to Theorem 1 (the policy is greedy HAP up to tie-breaking). With $\gamma = 0$ and $C > 0$, the *full* interior of Λ is kept and only the bound degrades. **Proof obligations:** P2 the positive-part handling; P3 interaction with $g_{\min}$ holds and the $g_{\max}$ override (bounded additive slack over a frame, as in Prop. 2); P4 composition with clearance loss (C1). Each proof is checked numerically (iletheory/sfsim.py) before it enters the paper.

Why CAS is better than the v2 filter.

1. The guarantee holds for every action, not only after a threshold is crossed.
2. There is no discontinuous override, so no chattering (P4c-11).
3. Every executed action is the policy's own, so no gradient masking is needed.
4. (γ, C) is an explicit, reportable knob trading freedom against the guarantee.
5. It composes with sensing error, which Com-MP's theory does not model.

Empirical questions (pre-registered): how much performance the constraint costs (H4), and the (γ, C) frontier.

C4. NEW — Storage-aware pressure and finite storage

Use the validated movement-resolution throttle $\mu_m^{\text{eff}} = \mu_m\,\varphi(h_{d(m)})$ with $\varphi(h) = \min(1, h/\theta)$ (P4c-1/P4c-7) inside the CAS pressure.

Proposition 3 (Easy part — to be proved)**.** *(a) If $h \geq \theta$ on every downstream link, storage-aware HAP equals HAP, so all C1–C3 results apply unchanged. (b) No phase that only discharges into full links ($h = 0$) can be preferred over a phase with positive pressure.*

Theorem target (risk: medium–high). Stability on a reduced region $\Lambda_\theta \subseteq \Lambda$ for a finite-buffer store-and-forward model with blocking, following the pressure-releasing analysis (Xiao et al. 2014) and PWBP (Li & Jabari 2019). **Fallback if not provable within the time box (3 weeks):** present the proposition plus a statistically powered deadlock-rate study (E4). The gap is explicitly open in the 2026 literature (PASCAL, VT-MP), so even a partial result is publishable.

C5. NEW — Data-grounded sensing model (GPU track)

Replace the assumed uniform confusion kernel with an *empirical* kernel and per-class miss rates measured from the released UVH-26 detectors on the UVH-26 validation split. Keep the single scalar η by interpolating between identity and the empirical kernel. Report robustness for both the synthetic grid and the real-calibrated condition.

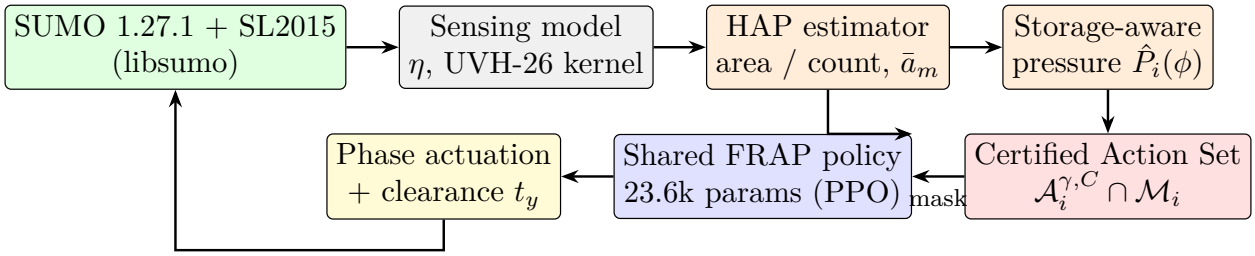
C6. Reviewer-proof evaluation

Fidelity-gated baselines including Com-MP and finite-storage MP, ≥3 training seeds, hierarchical statistics, a real-geometry Bengaluru corridor, and a powered deadlock test (Secs. 8–10).

Considered and deliberately excluded.

- **LLM-based controllers** (LLMLight, Traffic-R1). They conflict with the edge-deployable, guarantee-first story and would dilute it. Mention as related work only.
- **Large transformer policies.** The 23.6k-parameter shared policy is a selling point; keep it and report edge latency via ONNX on CPU.
- **A CityFlow port.** It cannot model sublane two-wheelers.

6.1 v3 architecture



7 Real-data grounding with UVH-26 (GPU track)

1. **Composition (cited fleet mix).** Map 14 classes to 5 model classes: 2W←two-wheeler (+bicycle reported separately); auto←three-wheeler; car←hatchback, sedan, SUV, MUV, van; bus←bus, mini-bus, tempo-traveller; truck←truck, LCV. Report shares by count and by box area, with bootstrap CIs over images. These are *in-view* (occupancy) shares, which is exactly the queue-composition notion the estimator needs (P1c-9). Flow shares differ; state this.
2. **Detector noise.** Run the released UVH-26 detectors on the validation split (on the GPU). Match predictions to ground truth (IoU ≥ 0.5) to obtain the 5×5 confusion matrix, per-class recall (dropout q_v) and a count-error distribution vs crowd density. Fit (θ, q) and the empirical kernel U_{emp}.
3. **Lemma 1 on real frames.** For each image, compare $\sum_{\text{boxes}} \tilde{a}/\bar{a}$ with $\sum_v \rho_v n_v$, where $\tilde{a}$ is the box area normalised by the image's median car-box area, a perspective proxy. Report the dispersion of this ratio under HCM and area weights. This is an *upper bound* on the field ε, because perspective variation inflates it. Per-camera calibration would remove the uniform part, which is free by Remark 3.
4. **Deliverables:** `vision/uvh26_*.py` scripts (no notebooks in the pipeline), numbers written to the config, and a data card with licence and citation.

Limits to state: 2D image boxes are not ground-plane area; no camera IDs; daytime frames only (06:00–18:00); February 2025 only.

8 Scenarios

S1 Arterial (keep). 5 signals, 400 m. Recalibrate capacity after the fix.

S2 Grid (keep). 4×4, 300 m. This is the spillback-prone case where HAP shows directional AQL and spillback gains (P1c-4).

S3 Bengaluru OSM-informed corridor (rebuild). Use the 5 verified Old Madras Road signals (`416419153`, `cluster_11165270540_...`, `442438288`, `343459489`, `1837297484`; span 5.7 km). Take real coordinates, spacing, approach lane counts and turn pockets from OSM and satellite imagery, but build the network *programmatically* (plain `nod/edg/con` XML) so that the 62% short single-lane segment problem never arises. Use a directional AM peak and the UVH-26 composition. Document it as "real geometry, abstracted topology", a common and accepted practice.

S4 Generalisation (optional, if time). A RESCO real-world SUMO network (Ingolstadt7 or Cologne8) with the Indian fleet injected, for zero-shot transfer.

Stress variants. v/c $\in \{0.85, 0.95, 1.05\}$, a 15-min surge, a lane-closure incident, 2W share ± 10 pp, composition dispersion $\{0, .25, .5, .75, 1\}$, and a bus-heavy approach.

9 Baselines and fidelity gate

Group	Methods	Status
Classical	Fixed-time (Webster on true demand), SOTL / SUMO actuated, MP (count), Efficient-MP, Advanced-MP (corrected), GreedyHAP, storage-aware GreedyHAP	exist
Finite-storage MP	**CABP** (Gregoire et al. 2015), **PWBP** (Li & Jabari 2019) or Hao et al. (2020)	new
Learned	IPPO, MPLight, PressLight, CoLight (*full* multi-head GAT, not single-hop), Advanced-MPLight (running demand as current-phase stay bonus, as published)	code exists, fix + retrain
Stability-preserving RL	**Com-MP** adapted (shared MAPPO mixes count-MP and HAP pressures), code at `github.com/bryan-zjc/Com-MP`	new
Optional SOTA	Unicorn (2025, code public) if the fidelity gate can be met within 1 week	optional
Ours (ablations)	HAPLight-RL with CAS / with v2 filter / unconstrained; (γ, C) sweep; storage-aware off; no neighbour info; no domain randomisation; per-intersection networks	new

Fidelity gate (must pass before any baseline enters a table).

1. Each learned baseline is trained on a *homogeneous-traffic standard benchmark* (RESCO `grid4x4` / `arterial4x4` in SUMO). Its learning curve and its ranking relative to MP and fixed-time must be consistent with published results. If not, the port is fixed or the method is dropped, with the reason recorded.
2. A **reward gate v2** is pre-registered *before* any v3 run. A reward passes if its episode return rank-correlates with $-$AQL (vehicle units) at $\rho \geq 0.9$, over ≥ 7 controllers and ≥ 3 seeds. Throughput is excluded when its spread is under 1% (the degenerate S2 criterion, P5-4). Each baseline trains with its *published* reward if that passes; otherwise with `total_queue`. This is declared a priori.
3. **Equal tuning budget.** Every learned method, ours included, gets the same small grid (learning rate $\{1, 3\} \times 10^{-4}$ $\times$ entropy $\{0.01, 0.003\}$) on S1, selected on validation seeds that are disjoint from test seeds. All methods share the same iteration budget, fixed as max(40, 2.5 $\times$ post-fix knee).

10 Pre-registered hypotheses and experiment matrix

Minimum practically relevant effect (MPE): 2% ATT, 5% AQL (unchanged from v2). Each experiment gets `experiments/EXX/prereg.md`, committed and git-tagged before launch. Refutations are reported.

ID	Hypothesis	Test / decision rule
H1	HAP−MP advantage (AQL in vehicle and weighted units, spillback time) grows with composition dispersion and is zero at zero dispersion.	Paired $n = 20$ per level, 5 levels, S2 and S3. Page / Spearman trend test. A null result is reported as a boundary of the contribution.
H2	HAPLight-RL (CAS) beats greedy HAP by ≥MPE on ATT in every scenario, and beats every learned baseline.	Wilcoxon + Holm per (scenario, metric) family, rank-biserial, hierarchical BCa.
H3	HAPLight-RL degrades more slowly with η than count-based methods, including $\eta \in \{0.8, 1.0\}$ (out of distribution) and the UVH-26 real-calibrated condition.	Slope of relative ATT vs η; bootstrap CI on the slope difference.
H4	CAS costs ≤MPE at nominal demand and lowers deadlock under stress.	TOST non-inferiority vs unconstrained RL (margin = MPE); McNemar on paired deadlock events.
H5	Weight-vector trade-off: area is better for the estimator, HCM for A5.	C1/C2 with ≥5 seeds plus UVH-26; $\varepsilon + \nu$ with CIs; control performance in a 2×2 (weights × estimator).
H6	Storage-aware pressure cuts the deadlock rate vs MP and HAP at ≤MPE nominal cost.	$n = 200$ paired stress episodes per cell (powered for 5% vs 1%), McNemar, Wilson CIs.
H7	A policy trained on S1+S2 transfers zero-shot to S3 (and S4) and retains its sign of advantage over greedy HAP.	Paired $n = 20$; report the retained fraction of the in-domain gain.

Exp.	Content	Design	Output
E0	Re-validation after the 8× fix	C1/C2 (5 seeds), capacity scales, $\bar{a}_m$ refit, B, storage threshold	corrected config + decision record
E1	Main comparison	S1–S3 × $\eta \in \{0, 0.6\}$ × all methods; 3–5 training seeds × 20 evaluation seeds	Table (main), effect sizes
E2	H1 dose–response	5 dispersion levels, paired, S2/S3	Figure + trend test
E3	Robustness	$\eta \in \{0, .2, .4, .6, .8, 1\}$ + UVH-26 kernel, S2/S3	Robustness curves
E4	Deadlock power study	v/c 0.95/1.05 + surge, 200 paired episodes per cell, 6 methods	Deadlock table
E5	Ablations	CAS / filter / none; (γ, C) grid; storage-aware off; neighbours off; domain randomisation off; IPPO; reward forms; $c_{\text{sw}} = 0$	Ablation table, (γ, C) frontier
E6	Weight vector (H5)	HCM vs area × count vs area estimator	2×2 table
E7	Fairness	per-class AWT, Gini, PWD; 2W penalty check	Fairness table
E8	Sensitivity	$\Delta t \in \{5, 10\}$, B or C, ±20% equivalency, 2W ±10 pp, bus occupancy	Sensitivity figure
E9	Generalisation	zero-shot S3/S4, unseen demand profiles	Transfer table
E10	Deployability	ONNX CPU latency, parameter count, sensing bytes per Δt	Text + table
T	Theorem checks	store-and-forward simulator: CAS bounded inside $(1-\gamma)\Lambda$, divergent outside; measured vs predicted bound	Appendix figure, CI test

Statistics.

- **Nested design.** Training seed ⊃ evaluation seed, so CIs use a *hierarchical bootstrap*: resample training seeds, then evaluation seeds.
- **Location and tests.** Medians are primary (heavy tails from deadlock episodes). Wilcoxon with Holm correction within each (scenario, metric) family; rank-biserial r; BCa 95% CIs.
- **Rare events.** Wilson CIs for rates and McNemar for paired deadlock.
- **Non-inferiority.** TOST.
- **Reporting.** `eval/stats.py` and `eval/report.py` emit every LaTeX table. No number is ever typed by hand.

11 Compute plan

Measured today (TraCI, system Python, OneDrive): one PPO iteration (8 parallel 3600 s episodes plus update) takes ≈430 s on S1 and ≈750 s on S2. Evaluation episodes take ≈220–250 s (S1) and ≈340 s (S2) each.

Expected after Phase A: libsumo, off OneDrive, one environment. The 1.5–2.5× speed-up must be *measured* before the budget below is locked.

Block (40 iterations, 3 training seeds)	current (h)	after Phase A, est. 2× (h)
Main learned set: 8 methods × S1+S2 × 3 seeds (S1 ≈5 h, S2 ≈8.5 h per run)	324	162
Ablations on S2: 6 variants × 3 seeds	153	77
S3 training: 4 methods × 3 seeds	72	36
Evaluation: all experiments, ≈7,500 episodes at 6 workers	85	43
Theory simulator, UVH-26 inference (GPU), re-validation	15	15
Total	**649** (≈27 days)	**333** (≈14 days)

Compute tiers.

Tier 1, laptop only, after Phase A. ≈14 days of continuous compute. This fits the 12-week plan only if the queue never idles.

Tier 2, plus RAM upgrade to 16–32 GB. 12–14 workers instead of 6–8, an extra ≈1.5×, giving ≈9–10 days. **Recommended.**

Tier 3, cloud or HPC burst for E1 and E5. A 32–64 vCPU instance, or the RGIPT cluster, running 4–6 runs concurrently. The sweep finishes in 3–4 days. Price it at the time of booking.

Cut order if over budget: (1) S4 and Unicorn; (2) ablations down to 2 seeds; (3) robustness grid on S2 only; (4) evaluation seeds 20→15. Never: training seeds below 3, or unequal budgets.

12 Repository blueprint and engineering practices

12.1 Location and layout

Code in `C:\dev\haplight` (NVMe, not synced). Bulk artefacts in `D:\haplight-data` via the environment variable `HAPLIGHT_DATA`. A private GitHub remote. OneDrive only receives weekly zipped snapshots. The current folder is frozen as `D:\haplight-archive\HAPLIGHT_v2_2026-11-04.zip` and its changelog is imported as `docs/history/`.

```
haplight/
  pyproject.toml  uv.lock  README.md  CLAUDE.md  CITATION.cff  LICENSE
  .pre-commit-config.yaml  .github/workflows/ci.yml  .gitignore
  configs/          base.yaml + scenario/*.yaml + experiment/*.yaml  (pydantic-validated)
  src/haplight/
    sim/            env.py, measurement.py, topology.py, backend.py (libsumo|traci)
    hap/            weights.py, estimators.py, pressure.py, storage.py
    sensing/        noise.py, kernels.py (synthetic + UVH-26 empirical)
    control/        classical.py (FT,SOTL,MP,EMP,AMP,CABP,PWBP,GreedyHAP), cas.py, filter.py
    rl/             policy.py, ppo.py, rollout.py, train.py
```

```
  baselines/       mplight.py, presslight.py, colight.py, amplight.py, ippo.py, commp.py
  theory/          sfsim.py  (fast store-and-forward simulator for theorem checks)
  eval/            runner.py, metrics.py, stats.py, report.py
  vision/          uvh26_composition.py, uvh26_confusion.py, uvh26_area.py
  utils/           seeding.py, manifest.py, io.py, log.py
scripts/           thin CLIs only (train, evaluate, calibrate, build_nets, report)
experiments/E00..E10/   prereg.md, config.yaml, analysis.py
nets/              s1, s2, s3_bengaluru (generated, with build provenance)
tests/unit  tests/integration  tests/regression
paper/             LaTeX; paper/generated/ (tables from report.py only)
docs/              plan/, decisions/ (ADR), STATUS.md, history/
```

12.2 Practices (each fixes a failure observed in v2)

Practice	Why (observed failure)
One `uv`-locked environment, Python 3.12, SUMO 1.27.1 everywhere	Two divergent environments; training silently ran traci 1.25 against the 1.27.1 binary.
libsumo backend + `test_backend_parity` (traci vs libsumo, same seed ⇒ identical trajectory)	Socket crash (13 Sep); 2–3× speed.
Fresh process per episode (`maxtasksperchild=1`) enforced by a structural test	State leakage across episodes, confirmed six times (P4c-14/15).
Ground-truth measurement gate in CI (`test_counts_match_ground_truth`)	The 8× double-counting bug survived because tests compared counts with counts.
Every run writes `manifest.json`: git SHA, dirty flag, config hash, seed, package versions, backend, measurement-pipeline version, host, wall-clock	Pre- and post-fix runs were compared by mistake (P5-3 retraction). Comparisons across pipeline versions are refused by code.
Results as parquet, one row per episode, plus a run index	Results scattered across JSON, CSV and stdout files.
Pre-registration file + git tag before each experiment	Threshold and gate decisions were made after seeing data and had to be retracted.
`docs/STATUS.md` updated at the end of every session; ADR decision records	This session started with no knowledge of where work stopped.
`CLAUDE.md` with the project rules (no paper edits during experiments, config is the single source of numbers, gates, cooldowns, never touch the user's processes)	Keeps AI development sessions consistent.
ruff (lint + format), mypy (gradual), pytest markers (`unit`, `sumo`, `slow`), pre-commit, GitHub Actions (unit + lint on every push; SUMO integration nightly)	Prevent regressions; the suite takes 16 min, so split it by marker.
Long runs: chain scripts with retry watchdog, cooldowns, resumable checkpoints, logs on D:	Laptop throttling, socket failures, OneDrive contention.

12.3 Migration from the v2 folder

1. Archive the folder (zip, read-only) and record SHA-256 checksums of the valid checkpoints.
2. Port modules into the package layout *without behaviour change*. The ported tests (84) must pass bit-identically on the same backend first, then on libsumo.
3. Port configs, nets and the valid evidence (P5-5 numbers) as regression fixtures.
4. Only then start v3 feature work (CAS, storage theory, vision, baselines).

13 Timeline and gates

Weeks	Dates (2026–27)	Work	Gate to pass
1	9–15 Nov	Repo + env + CI; backend parity; speed benchmark; port v2 code; download UVH-26 to D:; RAM upgrade ordered	84 tests green on libsumo; parity holds; speed-up measured
2	16–22 Nov	E0 re-validation; UVH-26 composition + confusion; CAS implementation + unit tests; theory drafts P1–P4; S3 rebuild started	E0 decision record; CAS tests; composition table
3	23–29 Nov	Store-and-forward theorem checks; HAPLight-RL-CAS pilot (S1, 3 seeds); CABP/PWBP; reward gate v2	Pilot beats greedy HAP; theorem check passes
4–5	30 Nov–13 Dec	Baseline fidelity gates on RESCO; Com-MP adaptation; equal HP tuning; S3 calibration; finite-storage theorem time box	All baselines gated; S3 loads, MP beats FT under stress
6–8	14 Dec–3 Jan	E1 + E5 training sweep (Tier 2 or 3 compute)	All checkpoints present with manifests
9	4–10 Jan	E2–E4, E6–E10 evaluations	All parquet present
10	11–17 Jan	`stats.py/report.py`; results review; narrative decision (author review)	Every table generated
11–12	18–31 Jan	Paper v3 rewrite (from the decision log + generated tables); internal review; arXiv	0 placeholders; every proof independently re-checked
13	early Feb 2027	Submit (IEEE T-ITS or TR-C)	—

14 Risk register

Risk	Likelihood	Mitigation
H1 null: HAP ≈ MP on delay	high	Already likely. Reframe HAP as the correct units for stability (A5) and sensing (Lemma 1). Gains are claimed from CAS-RL and storage-awareness. Report H1 as a boundary result.
Finite-storage theorem not provable in time	med–high	3-week time box, then fall back to proposition + powered study. The gap is open in the 2026 literature.
Overlap with Com-MP or another preprint appears	medium	Cite, compare as a baseline, stress the differentiators (vehicle heterogeneity, sensing theory, real data). Post to arXiv as soon as E1 is done.
Compute overrun	medium	Tiers 2/3; cut order; shared iteration budget.
Baseline-port criticism	medium	Fidelity gate on RESCO; published-reward variants; equal tuning budget.
CAS too restrictive (RL ≈ greedy)	medium	The (γ, C) frontier is itself a result. Choose the operating point on validation seeds only.
Backend non-determinism / crashes	low–med	Parity test; retry watchdog; fresh process per episode.
Another measurement bug	low	Ground-truth gate in CI; pipeline version in manifests.
Loss of session context	medium	`STATUS.md`, ADRs and `CLAUDE.md`; commit and push every session.
UVH-26 analysis challenged (2D boxes, no camera IDs)	medium	Present as an upper-bound/approximate validation with the limits stated.

15 Immediate next steps (first two weeks)

Decisions taken on 4 November 2026

1. **Repository:** code in `C:\dev\haplight` (https://github.com/jarvisxys/haplight), data in `D:\haplight-data`. The v2 folder is archived in `D:\haplight-data\archive`.
2. **No RAM upgrade.** The laptop is for development, tests and small pilots (Tier 1).
3. **Heavy compute** (E1, E4, E5) runs on the college server with an NVIDIA RTX PRO 6000 Blackwell (96 GB); this replaces the cloud burst of Tier 3. SUMO is CPU-bound, so throughput depends on the server's CPU core count and RAM (to be confirmed). The GPU serves the UVH-26 vision track and needs a PyTorch build with CUDA 12.8 or later.
4. **Main contribution:** the Certified Action Set replaces the safety filter, which stays as an ablation.
5. **Authorship:** single author for now (Aditya Singh); co-authors to be added later.

1. **Decide**: repository location (proposed `C:\dev` + `D:\haplight-data`), the GitHub remote, the RAM upgrade, and whether a cloud or HPC burst is available.
2. Snapshot the current folder (git-init plus zip archive). Do *not* resume `_phase5_retrain4.sh`; v3 supersedes it.
3. Create the v3 repository and `uv` environment. Port the code and tests. Run the parity test and the speed benchmark.
4. Run E0 re-validation (C1/C2 seeded, capacity, $\bar{a}_m$, filter B).
5. Download UVH-26 (validation split + all annotation JSON) to D:. Compute the composition and the detector confusion matrix on the GPU.
6. Implement `control/cas.py` and its tests (argmax $\in$ CAS; CAS(0,0) $\equiv$ greedy; mask never empty) and `theory/sfsim.py`.
7. Draft Theorem 2 (CAS) and Proposition 3 (storage) with proof obligations P1–P4.
8. Pilot HAPLight-RL-CAS on S1 (3 seeds, 40 iterations) against greedy HAP.
9. Write and tag `experiments/E01/prereg.md` and reward gate v2.
10. Start the S3 OSM-informed corridor build.

A Key references (verified during this audit)

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B Glossary of project-specific terms

8× bug (P5-2) Context subscriptions counted every vehicle once per overlapping edge context. Fixed by deduplicating by vehicle ID; guarded by a ground-truth test.

S1 / S2 / S3 5-signal arterial / 4×4 grid / Bengaluru corridor.

η Composite sensing-noise level in $[0, 1]$. Training draws $\eta \sim \mathcal{U}[0, 0.6]$.

`total_queue` Reward $-\frac{1}{x_{\text{ref}}} \sum_{m \in \mathcal{M}_i} x_m$ (true channel), validated by the reward gate.

Gridlock / deadlock Binary per episode: SUMO still has pending (un-inserted) vehicles at truncation, `env.pending_count()` > 0 (the evaluator's `GRIDLOCK_KEY`). The diagnostic signature is worst-movement storage occupancy pinned above 100%. v3 should pre-register this definition, plus a continuous severity measure.